Runway Surface Condition Report

Runway survey — UAV imagery, ~1.25 mm GSD · generated 2026-06-15

136 m

surveyed length

0.76%

crack density (SAM2 true area)

236 m

total crack length

267

cracks measured

207/274

crack-bearing frames

0.10%

vegetation coverage

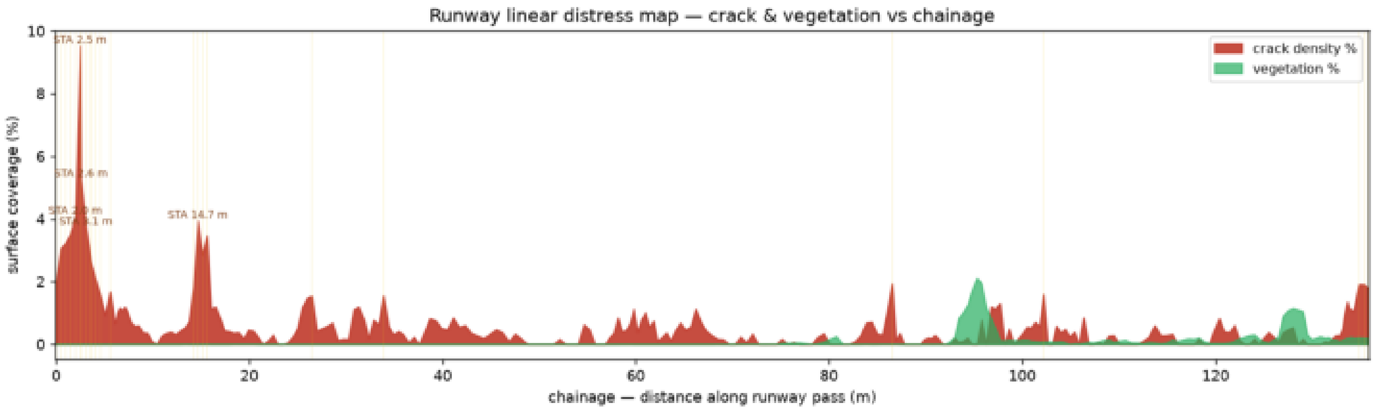
Automated surface screening of the runway from a continuous drone pass (custom YOLOv8 detection; crack masks refined with SAM2). Findings are referenced by chainage (distance along the pass) and summarised as crack density and vegetation coverage with hot zones to prioritise maintenance, consistent with routine pavement monitoring under ICAO Annex 14 / CAA CAP 168 and the ASTM D5340 distress framework.

Over 136 m surveyed the model maps 267 cracks (236 m total length) across 207 frames, a true crack density of 0.76% of imaged surface, and vegetation in 85 frames.

Scope & limitations: crack density is measured from SAM2 masks (relative index for prioritisation/trend). Per-crack WIDTH, ASTM severity grading and a numeric PCI are NOT reported — motion blur and grooved-runway texture make width unreliable on this survey (see Limitations). Precision/recall are not quantified (no verified ground truth). FOD, rubber deposits and marking condition are out of scope here (see the ASTM coverage matrix).

Spatial distress map (by chainage)

Distance along the runway pass, recovered by frame registration.

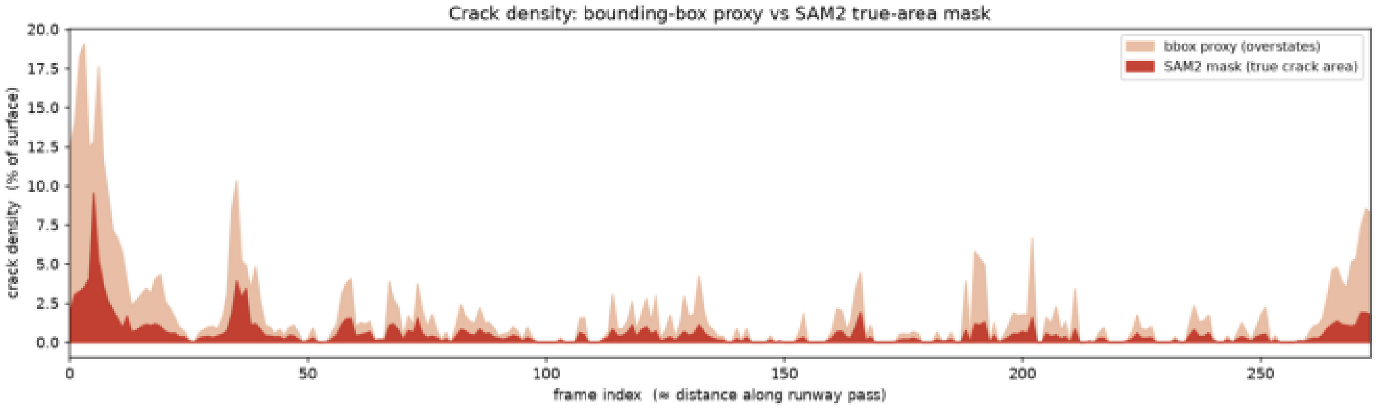


Priority crack hot zones (station · density):

STA	2.5 m	9.53%
STA	2.6 m	5.26%
STA	2.0 m	4.08%
STA	14.7 m	3.94%
STA	3.1 m	3.74%
STA	1.5 m	3.51%
STA	15.6 m	3.46%
STA	1.0 m	3.25%

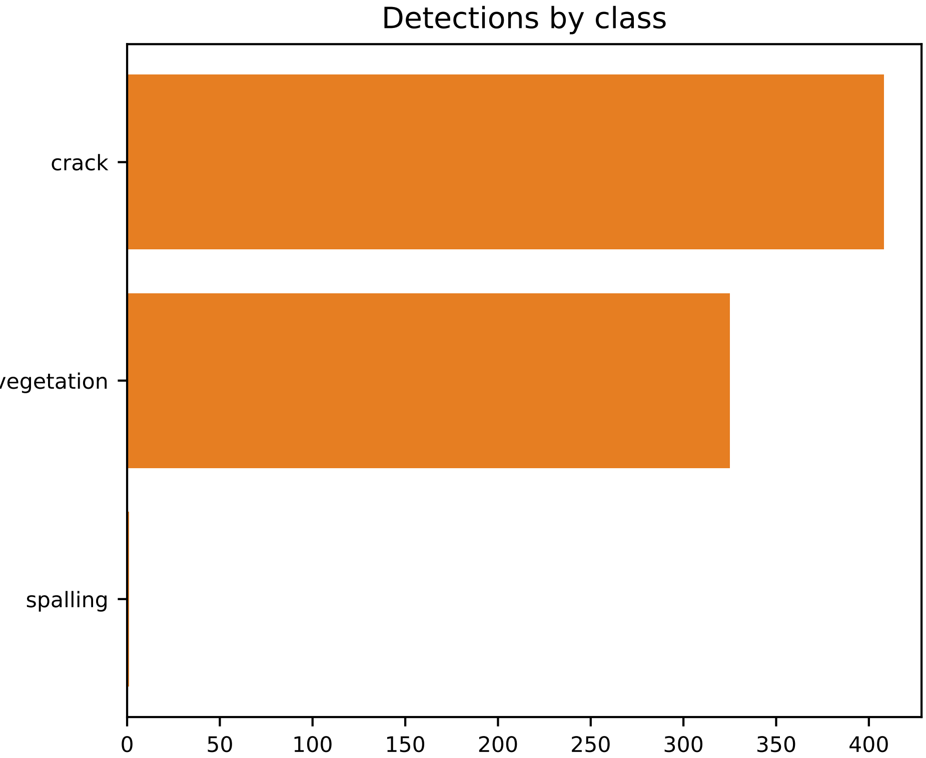
Crack density — measurement

Bounding-box proxy vs SAM2 true-area mask.



Bounding boxes overstate crack area ~3x; SAM2 masks give the true crack-covered area. Both preserve the same hot-zone pattern — only the magnitude is corrected.

Feature inventory

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Cracks
  detections: 408
  measured cracks: 267
  total length: 236 m
  type mix: {'longitudinal': 262, 'trans

Vegetation
  coverage: 0.10% overall
  present in 85/274 frames
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Crack type is from skeleton orientation (geometry is robust);
vegetation from a green colour threshold (separable from asphalt).

ASTM D5340 coverage matrix

What this survey assesses, and what needs more.

item	status	note
Cracking (long./trans.)	ASSESSED	density, length, location, type
Vegetation encroachment	ASSESSED	coverage % + hot zones (operational)
Patching	PARTIAL	detector class present; not validated here
Crack width / severity / PCI	NOT REPORTED	blur + groove texture (see Limitations)
FOD	OUT OF SCOPE	needs a trained FOD model (FOD-A)
Rubber deposits	OUT OF SCOPE	needs TDZ location + friction data
Raveling / weathering	OUT OF SCOPE	texture; unreliable on blurred RGB
Marking condition	OUT OF SCOPE	colour threshold catches shoulder/verge
Ponding / drainage	N/A	dry survey

Crack hot zones — SAM2 masks

red = crack mask · yellow = detection box

frame_000006.jpg — STA 2.5 m — 9.53% density

frame_000007.jpg — STA 2.6 m — 5.26% density

Crack hot zones — SAM2 masks

red = crack mask · yellow = detection box

frame_000005.jpg — STA 2.0 m — 4.08% density

frame_000036.jpg — STA 14.7 m — 3.94% density

Measurement limitations & upgrade path


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Why per-crack width, ASTM severity and PCI are not reported:

  - Motion blur and grooved-runway texture (the same challenges
    noted in published runway-crack studies) blur the thin crack
    edge needed for width.
  - Vanilla SAM2 segments a broad region, not the 1-3 px crack
    line, so mask width overstates (~46 mm median — not physical).
  - A dark-line method collapses to a near-constant ~5 mm, i.e. it
    measures the groove/texture scale, not real crack width.

What IS reliable here: crack density & hot zones (prioritisation),
crack count/location/length, crack type mix, vegetation coverage,
and chainage referencing.

Upgrade path to certified-style width + ASTM severity + PCI:
  - capture deblurred / slower (stop-and-stare) imagery, and/or
  - use a crack-fine-tuned segmenter (CrackSAM) instead of vanilla
    SAM2; then width-based severity and PCI become trustworthy.

Other notes: precision/recall need human-verified ground truth;
a single video pass yields a strip, not a full-width orthomosaic.
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Distress list — cracks by chainage

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id	STA(m)	frame	type	len(m)
3	0.0	frame_000001.jpg	longitudinal	2.83
4	0.5	frame_000002.jpg	longitudinal	1.44
7	0.5	frame_000002.jpg	longitudinal	2.90
8	1.0	frame_000003.jpg	longitudinal	1.47
9	1.0	frame_000003.jpg	longitudinal	2.78
10	1.0	frame_000003.jpg	longitudinal	2.80
11	1.0	frame_000003.jpg	longitudinal	3.02
13	1.0	frame_000003.jpg	longitudinal	0.60
15	1.5	frame_000004.jpg	longitudinal	3.26
18	1.5	frame_000004.jpg	longitudinal	2.92
19	1.5	frame_000004.jpg	longitudinal	2.06
20	2.0	frame_000005.jpg	longitudinal	3.29
21	2.0	frame_000005.jpg	longitudinal	1.60
22	2.0	frame_000005.jpg	longitudinal	0.63
30	2.6	frame_000007.jpg	longitudinal	3.24
36	3.6	frame_000009.jpg	longitudinal	1.51
39	4.1	frame_000010.jpg	longitudinal	1.53
43	4.6	frame_000011.jpg	longitudinal	1.96
44	4.6	frame_000011.jpg	longitudinal	1.88
47	5.1	frame_000012.jpg	longitudinal	1.12
48	5.1	frame_000012.jpg	longitudinal	1.02
50	5.6	frame_000013.jpg	longitudinal	0.62
51	5.6	frame_000013.jpg	longitudinal	3.00
55	6.2	frame_000015.jpg	longitudinal	0.37
58	6.4	frame_000016.jpg	longitudinal	0.41
59	6.6	frame_000017.jpg	longitudinal	0.66
60	6.6	frame_000017.jpg	longitudinal	1.63
63	6.8	frame_000018.jpg	longitudinal	0.62
65	7.1	frame_000019.jpg	longitudinal	0.59
68	7.1	frame_000019.jpg	longitudinal	0.47
69	7.4	frame_000020.jpg	longitudinal	0.60
76	7.8	frame_000021.jpg	longitudinal	0.61
83	9.9	frame_000026.jpg	longitudinal	0.30
84	10.9	frame_000028.jpg	longitudinal	0.33
85	10.9	frame_000028.jpg	longitudinal	0.44
86	11.4	frame_000029.jpg	longitudinal	0.89
87	11.9	frame_000030.jpg	longitudinal	0.59
88	11.9	frame_000030.jpg	longitudinal	0.45
89	12.4	frame_000031.jpg	longitudinal	0.81
91	13.3	frame_000033.jpg	longitudinal	1.33
92	13.8	frame_000034.jpg	longitudinal	1.81
100	15.6	frame_000038.jpg	longitudinal	0.42
102	16.1	frame_000039.jpg	longitudinal	0.90
104	16.5	frame_000040.jpg	longitudinal	1.02
107	17.0	frame_000041.jpg	longitudinal	0.92
115	19.4	frame_000046.jpg	longitudinal	0.49
116	19.9	frame_000047.jpg	longitudinal	0.17
118	20.4	frame_000048.jpg	longitudinal	0.21
119	20.4	frame_000048.jpg	longitudinal	0.23
120	20.4	frame_000048.jpg	longitudinal	0.78
121	20.9	frame_000049.jpg	longitudinal	0.71
122	21.9	frame_000051.jpg	longitudinal	0.41
123	22.5	frame_000052.jpg	longitudinal	0.92
124	24.0	frame_000055.jpg	longitudinal	0.14
125	24.5	frame_000056.jpg	longitudinal	0.68
126	25.0	frame_000057.jpg	longitudinal	0.19
129	25.0	frame_000057.jpg	longitudinal	0.77
131	25.5	frame_000058.jpg	longitudinal	0.81
133	25.5	frame_000058.jpg	longitudinal	0.41
134	26.0	frame_000059.jpg	longitudinal	0.83
136	26.0	frame_000059.jpg	longitudinal	0.46
137	26.0	frame_000059.jpg	longitudinal	0.50
139	26.5	frame_000060.jpg	longitudinal	0.80
142	26.5	frame_000060.jpg	longitudinal	1.38
143	27.0	frame_000061.jpg	longitudinal	0.34
144	27.0	frame_000061.jpg	longitudinal	0.24
145	27.0	frame_000061.jpg	longitudinal	0.54
147	27.5	frame_000062.jpg	longitudinal	0.77
148	27.5	frame_000062.jpg	longitudinal	0.54
149	28.1	frame_000063.jpg	longitudinal	0.58
154	29.2	frame_000065.jpg	longitudinal	0.35
155	29.7	frame_000066.jpg	longitudinal	0.41
156	30.3	frame_000067.jpg	longitudinal	0.41
158	30.8	frame_000068.jpg	longitudinal	0.58
162	31.3	frame_000069.jpg	longitudinal	0.48
163	31.3	frame_000069.jpg	longitudinal	0.59
164	31.9	frame_000070.jpg	longitudinal	0.59
165	31.9	frame_000070.jpg	longitudinal	0.99
166	31.9	frame_000070.jpg	longitudinal	0.55
167	32.4	frame_000071.jpg	longitudinal	0.51
168	32.9	frame_000072.jpg	longitudinal	0.95
171	33.4	frame_000073.jpg	longitudinal	0.48
173	33.4	frame_000073.jpg	longitudinal	0.25
174	33.9	frame_000074.jpg	longitudinal	0.55
176	34.4	frame_000075.jpg	longitudinal	1.33
177	34.9	frame_000076.jpg	longitudinal	0.81
178	35.5	frame_000077.jpg	longitudinal	1.15
179	36.0	frame_000078.jpg	longitudinal	0.73
180	36.5	frame_000079.jpg	longitudinal	0.17
181	37.0	frame_000080.jpg	longitudinal	0.64
182	38.1	frame_000082.jpg	longitudinal	0.72
183	38.1	frame_000082.jpg	longitudinal	0.66

Distress list — cracks by chainage

id	STA(m)	frame	type	len(m)
184	38.7	frame_000083.jpg	longitudinal	0.60
187	39.3	frame_000084.jpg	longitudinal	0.62
188	39.3	frame_000084.jpg	longitudinal	0.42
189	39.9	frame_000085.jpg	longitudinal	0.96
190	39.9	frame_000085.jpg	longitudinal	0.54
191	40.5	frame_000086.jpg	longitudinal	0.28
192	40.5	frame_000086.jpg	longitudinal	1.12
194	41.1	frame_000087.jpg	longitudinal	1.03
195	41.1	frame_000087.jpg	longitudinal	0.11
197	41.7	frame_000088.jpg	longitudinal	1.07
198	42.3	frame_000089.jpg	longitudinal	0.52
199	42.3	frame_000089.jpg	longitudinal	0.72
200	42.3	frame_000089.jpg	longitudinal	0.32
201	43.0	frame_000090.jpg	longitudinal	0.48
202	43.0	frame_000090.jpg	longitudinal	0.67
204	43.6	frame_000091.jpg	longitudinal	0.17
205	44.2	frame_000092.jpg	longitudinal	0.68
208	45.5	frame_000094.jpg	longitudinal	0.48
210	46.2	frame_000095.jpg	longitudinal	0.32
211	46.9	frame_000096.jpg	longitudinal	0.29
212	47.5	frame_000097.jpg	longitudinal	1.00
213	48.2	frame_000098.jpg	longitudinal	0.42
215	54.6	frame_000108.jpg	longitudinal	0.70
217	55.2	frame_000109.jpg	longitudinal	1.27
218	57.3	frame_000113.jpg	longitudinal	0.20
222	58.1	frame_000115.jpg	longitudinal	1.54
224	58.6	frame_000116.jpg	longitudinal	0.35
226	59.0	frame_000117.jpg	longitudinal	0.31
229	59.4	frame_000118.jpg	longitudinal	0.34
232	59.8	frame_000119.jpg	longitudinal	0.91
233	59.8	frame_000119.jpg	longitudinal	0.33
236	60.6	frame_000121.jpg	longitudinal	0.33
237	60.6	frame_000121.jpg	longitudinal	0.44
240	61.0	frame_000122.jpg	longitudinal	0.83
241	61.0	frame_000122.jpg	longitudinal	0.85
243	61.4	frame_000123.jpg	longitudinal	0.49
245	61.8	frame_000124.jpg	longitudinal	1.57
246	62.3	frame_000125.jpg	longitudinal	0.40
247	62.7	frame_000126.jpg	longitudinal	0.84
248	63.2	frame_000127.jpg	longitudinal	1.27
249	63.6	frame_000128.jpg	longitudinal	0.48
250	64.1	frame_000129.jpg	longitudinal	0.94
251	64.1	frame_000129.jpg	longitudinal	0.36
252	64.6	frame_000130.jpg	longitudinal	0.87
253	64.6	frame_000130.jpg	longitudinal	1.43
254	65.1	frame_000131.jpg	longitudinal	1.26
255	65.7	frame_000132.jpg	longitudinal	1.22
256	65.7	frame_000132.jpg	longitudinal	0.25
257	66.2	frame_000133.jpg	longitudinal	0.79
258	66.2	frame_000133.jpg	longitudinal	1.21
259	66.2	frame_000133.jpg	longitudinal	1.23
261	66.8	frame_000134.jpg	longitudinal	0.86
262	67.3	frame_000135.jpg	longitudinal	0.82
263	67.3	frame_000135.jpg	longitudinal	0.39
264	67.9	frame_000136.jpg	longitudinal	0.78
265	68.5	frame_000137.jpg	longitudinal	0.43
266	69.1	frame_000138.jpg	longitudinal	0.55
267	70.8	frame_000141.jpg	longitudinal	0.84
268	71.5	frame_000142.jpg	longitudinal	0.22
269	72.1	frame_000143.jpg	longitudinal	0.92
270	75.1	frame_000148.jpg	longitudinal	0.57
271	76.3	frame_000150.jpg	longitudinal	0.19
272	78.8	frame_000154.jpg	longitudinal	0.64
273	79.5	frame_000155.jpg	longitudinal	1.25
274	82.7	frame_000160.jpg	longitudinal	0.27
275	83.3	frame_000161.jpg	longitudinal	0.76
276	83.9	frame_000162.jpg	longitudinal	1.03
277	83.9	frame_000162.jpg	longitudinal	0.73
278	84.4	frame_000163.jpg	longitudinal	0.67
279	84.4	frame_000163.jpg	transverse	0.49
280	84.4	frame_000163.jpg	longitudinal	0.62
282	85.0	frame_000164.jpg	longitudinal	0.82
283	85.5	frame_000165.jpg	transverse	0.51
284	85.5	frame_000165.jpg	longitudinal	0.34
285	85.5	frame_000165.jpg	transverse	0.56
286	86.0	frame_000166.jpg	transverse	0.59
289	86.5	frame_000167.jpg	longitudinal	0.69
291	86.5	frame_000167.jpg	transverse	0.64
293	86.9	frame_000168.jpg	longitudinal	0.44
294	87.3	frame_000169.jpg	longitudinal	0.92
295	89.9	frame_000175.jpg	longitudinal	0.32
297	90.2	frame_000176.jpg	longitudinal	0.69
298	90.6	frame_000177.jpg	longitudinal	0.63
299	91.0	frame_000178.jpg	longitudinal	0.25
300	91.0	frame_000178.jpg	longitudinal	0.65
301	91.4	frame_000179.jpg	longitudinal	0.61
302	93.0	frame_000183.jpg	longitudinal	0.67
303	93.4	frame_000184.jpg	longitudinal	0.26
305	94.3	frame_000186.jpg	longitudinal	0.60
306	95.8	frame_000189.jpg	longitudinal	2.06
308	97.2	frame_000192.jpg	longitudinal	2.30
309	97.2	frame_000192.jpg	longitudinal	0.56

Distress list — cracks by chainage

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id	STA(m)	frame	type	len(m)
311	97.7	frame_000193.jpg	longitudinal	1.17
313	99.6	frame_000197.jpg	longitudinal	0.53
314	100.0	frame_000198.jpg	longitudinal	0.97
315	100.5	frame_000199.jpg	longitudinal	1.59
316	100.9	frame_000200.jpg	longitudinal	1.38
322	103.5	frame_000206.jpg	longitudinal	0.66
324	104.0	frame_000207.jpg	longitudinal	1.10
325	104.5	frame_000208.jpg	longitudinal	1.54
326	104.9	frame_000209.jpg	longitudinal	0.76
327	105.4	frame_000210.jpg	longitudinal	1.20
328	106.3	frame_000212.jpg	longitudinal	2.28
329	107.9	frame_000215.jpg	longitudinal	0.21
330	108.9	frame_000217.jpg	longitudinal	0.52
331	108.9	frame_000217.jpg	longitudinal	0.32
332	109.5	frame_000218.jpg	longitudinal	0.86
333	112.5	frame_000223.jpg	longitudinal	0.31
334	113.1	frame_000224.jpg	longitudinal	0.81
335	113.7	frame_000225.jpg	longitudinal	0.82
336	113.7	frame_000225.jpg	longitudinal	0.90
337	114.3	frame_000226.jpg	longitudinal	0.82
338	114.9	frame_000227.jpg	longitudinal	0.73
339	115.5	frame_000228.jpg	longitudinal	0.81
340	118.3	frame_000233.jpg	longitudinal	0.54
341	119.8	frame_000236.jpg	longitudinal	0.78
342	119.8	frame_000236.jpg	longitudinal	0.48
343	120.4	frame_000237.jpg	longitudinal	1.01
345	120.9	frame_000238.jpg	longitudinal	1.10
346	121.5	frame_000239.jpg	longitudinal	1.09
347	122.0	frame_000240.jpg	longitudinal	0.90
348	122.0	frame_000240.jpg	longitudinal	0.75
349	122.5	frame_000241.jpg	longitudinal	0.38
350	124.0	frame_000244.jpg	longitudinal	0.52
351	125.0	frame_000246.jpg	longitudinal	0.59
352	125.5	frame_000247.jpg	longitudinal	1.04
353	126.0	frame_000248.jpg	longitudinal	0.76
354	126.5	frame_000249.jpg	longitudinal	0.29
355	127.0	frame_000250.jpg	longitudinal	0.80
356	127.0	frame_000250.jpg	longitudinal	0.35
357	127.5	frame_000251.jpg	longitudinal	1.30
358	128.0	frame_000252.jpg	longitudinal	1.51
359	129.0	frame_000254.jpg	longitudinal	0.58
360	131.3	frame_000259.jpg	longitudinal	0.33
361	131.7	frame_000260.jpg	longitudinal	0.26
362	132.1	frame_000261.jpg	longitudinal	0.61
363	132.5	frame_000262.jpg	longitudinal	1.01
364	132.8	frame_000263.jpg	longitudinal	0.94

id	STA(m)	frame	type	len(m)
365	133.1	frame_000264.jpg	longitudinal	0.96
366	133.1	frame_000264.jpg	longitudinal	0.50
367	133.3	frame_000265.jpg	longitudinal	0.76
368	133.3	frame_000265.jpg	longitudinal	0.96
369	133.3	frame_000265.jpg	longitudinal	0.88
370	133.5	frame_000266.jpg	longitudinal	0.91
371	133.5	frame_000266.jpg	longitudinal	0.79
372	133.5	frame_000266.jpg	longitudinal	1.60
373	133.7	frame_000267.jpg	longitudinal	1.07
374	133.7	frame_000267.jpg	longitudinal	0.63
375	133.7	frame_000267.jpg	longitudinal	1.50
376	133.7	frame_000267.jpg	longitudinal	0.22
379	133.8	frame_000268.jpg	longitudinal	1.19
381	133.8	frame_000268.jpg	longitudinal	1.35
382	133.8	frame_000268.jpg	longitudinal	0.45
384	134.0	frame_000269.jpg	longitudinal	1.31
385	134.0	frame_000269.jpg	longitudinal	1.08
386	134.0	frame_000269.jpg	longitudinal	0.25
388	134.2	frame_000270.jpg	longitudinal	0.90
389	134.2	frame_000270.jpg	longitudinal	1.22
390	134.2	frame_000270.jpg	longitudinal	1.57
391	134.4	frame_000271.jpg	longitudinal	0.84
393	134.4	frame_000271.jpg	longitudinal	1.68
394	134.4	frame_000271.jpg	longitudinal	1.38
395	134.4	frame_000271.jpg	longitudinal	0.19
397	134.9	frame_000272.jpg	longitudinal	1.76
398	134.9	frame_000272.jpg	longitudinal	2.64
399	134.9	frame_000272.jpg	longitudinal	0.78
400	135.3	frame_000273.jpg	longitudinal	0.86
401	135.3	frame_000273.jpg	longitudinal	1.71
402	135.3	frame_000273.jpg	longitudinal	2.65
403	135.3	frame_000273.jpg	longitudinal	1.45
404	135.8	frame_000274.jpg	longitudinal	0.82
405	135.8	frame_000274.jpg	longitudinal	1.66
406	135.8	frame_000274.jpg	longitudinal	2.58
407	135.8	frame_000274.jpg	longitudinal	1.42
408	135.8	frame_000274.jpg	longitudinal	0.19